

JOSHUA OFORI BOATENG

Center for Wireless, Communities and Innovation (WiCI)
Department of Electrical and Computer Engineering, Iowa State University
391C Durham Center, 613 Morrill Road, Ames, Iowa 50011, USA

✉ jboateng@iastate.edu [github](https://github.com/jboateng1) [Google Scholar](https://scholar.google.com/citations?user=jboateng1) [ORCID](https://orcid.org/0000-0001-9452-9452)

Research Summary

A researcher in wireless systems dedicated to advancing open, AI-native, and hardware-accelerated 6G Radio Access Networks (RANs) through practical experimentation and deployment. My overarching goal is to *democratize* next-generation wireless technology by ensuring that advanced networks are affordable, reliable, and accessible to rural communities, agricultural sectors, and underserved populations. My research concentrates on three interconnected areas: (i) open, software-defined RAN architectures designed to streamline network deployment and customization; (ii) AI-native RAN systems that seamlessly integrate AI workloads and baseband processing within shared, hardware-accelerated infrastructures; and (iii) large-scale, reproducible field experimentation aimed at transforming research prototypes into cutting-edge commercial systems.

Research Interests

Open, Programmable, and Intelligent NextG Radio Access Networks (RAN): Open RAN architectures, open-source 5G/6G protocol stacks (OCUDU, Duranta (OAI)), spectrum slicing, virtualized and multi-tenant software-defined radio (SDR) infrastructure.

AI-RAN and Hardware-Accelerated RAN: CUDA-accelerated physical layer (PHY/L1) (OCUDU, NVIDIA Aerial), and co-location of AI/ML and RAN workloads on shared accelerated infrastructure.

Wireless Testbeds, Living Labs, and Measurement: Large-scale experimental 5G/6G platforms, real-world network measurement, and reproducible over-the-air experimentation for rural and agricultural broadband connectivity.

Predictable URLLC and Deterministic Wireless: Measurement-driven latency modeling and system-level tuning for predictable low-latency, high-reliability 5G/6G in precision agriculture, IoT, and rural broadband.

Education

- Doctor of Philosophy (Ph.D.) in Computer Engineering** **May 2021 – May 2026**
Iowa State University *Ames, Iowa, USA*
 MAJOR: Computing and Networking Systems *CGPA: 3.80/4.00*
 DISSERTATION: Toward Democratizing 5G: Enabling Open, Reliable and Scalable Wireless Systems for Use and Innovation
 SUPERVISOR: [Prof. Hongwei Zhang](#)
- Bachelor of Science (B.Sc.) in Telecommunication Engineering** **Sep 2015 – Jun 2019**
Kwame Nkrumah University of Science and Technology (KNUST) *Kumasi, Ghana*
 PROJECT: Performance Analysis of Variable Pulse Position Modulation (VPPM) in Li-Fi Systems *CGPA: 3.98/4.00*
 DISTINCTION: Valedictorian (Overall Best Graduating Student)
 SUPERVISOR: [Prof. Jerry John Kponyo](#)

Experience

- Iowa State University – Center for Wireless, Communities and Innovation (WiCI)** **Jul 2026 – Present**
Research Assistant Professor, Department of Electrical and Computer Engineering *Ames, Iowa, USA*
- Building and evaluating GPU-accelerated RAN on the ARA platform with [OCUDU](#) and [NVIDIA Aerial L1 \(cuPHY\)](#), including Aerial L1 beneath OCUDU L2/L3 via SCF FAPI and OCUDU's in-tree CUDA PHY backend, as a substrate for AI-native RAN functions (neural receivers, ML-based channel estimation).
 - Co-designed the [KaiAir](#) interference-control and real-time scheduling algorithms in OCUDU as an open-source contribution; demonstrated a live OCUDU 5G gNB serving a [Linux Foundation Duranta](#) UE (formerly OAI) on a scouting drone, streaming crop imagery to edge AI for weed detection [**MERIF 2026 Best Demo Award**].
 - 6G systems developer on [AgSlicing](#) (NSF VINES Track 2, \$4.65M), a five-organization effort with UVA, Skylark Wireless, Salin 247, and Federated Wireless to deliver TRL-8 predictable RAN and spectrum slicing for precision agriculture on ARA.

- Represent Iowa State in the [AI-RAN Alliance](#) (*AI-and-RAN working group*) and the [OCUDU Technical Steering Committee](#) on AI-RAN and Hardware-Acceleration working groups, shaping open and AI-native RANs.
- Served as Demo and Poster Chair of [AgWireless'26](#) and organizer of the NSF AgRuralG Workshop, co-located with the U.S. [Farm Progress Show](#).

Microsoft Research (MSR)

Research Intern, Networking Research Group

Jan 2024 – May 2024

Redmond, Washington, USA

- Designed and implemented an end-to-end GEO-based 5G Non-Terrestrial Network (NTN) testbed using the OpenAirInterface 5G stack (gNB, UE, and core), RF simulator, and USRP SDRs to evaluate satellite-integrated 5G connectivity under GEO-scale propagation delay.

Iowa State University – Center for Wireless, Communities and Innovation (WiCI)

Graduate Research Assistant

May 2021 – May 2026

Ames, Iowa, USA

- Led the design, deployment, and operation of the 5G and Open RAN infrastructure of the [ARA Wireless Living Lab](#) (NSF PAWR, \$16M), a 60+ km rural testbed now serving **170+ researchers** across the U.S.
- Built [AraSDR](#), a first-of-its-kind outdoor, end-to-end fully programmable 5G SA living lab (NI N320 gNB and B210 UE SDRs, custom PA/LNA RF front-ends, containerized OpenAirInterface and srsRAN) with a remote experiment-control framework for reproducible experiments (IEEE ICC 2024).
- Designed [AraRACH](#), a slot-based 5G random-access scheme that achieved reliable connectivity over **1 mile**, the longest real-world range reported for open-source 5G stacks on USRP SDRs, and raised random-access success to **90–100%** in obstructed industrial settings [[IEEE NetSoft 2025 Best Paper Award](#)].
- Prototyped end-to-end O-RAN with OCUDU (formerly srsRAN), Duranta (formerly OpenAirInterface (OAI), and the [FlexRIC](#) near-RT RIC; integrated an open-source 5G core with commercial RAN (IEEE MILCOM 2025); co-developed AraMIMO [[ACM WiNTECH 2023 Best Paper Award](#)] and the ARA PAWR demo [[MERIF 2024 Best Demo Award](#)].
- Mentored NSF REU undergraduate researchers (2024, 2025) and onboarded external ARA users through the ARA User Program.

Vodafone Ghana

Robotic Process Automation (RPA) Developer

Sep 2020 – Apr 2021

Accra, Ghana

- Developed and deployed RPA workflows for customer onboarding and data processing, increasing operational efficiency by **50%**; served as Scrum Master coordinating sprint planning, stand-ups, and backlog delivery.

Kwame Nkrumah University of Science and Technology

Teaching and Research Assistant, Department of Telecommunication Engineering

Aug 2019 – Sep 2020

Kumasi, Ghana

- Taught digital signal processing, computer networking, and project design while conducting wireless and IoT security research that produced *two peer-reviewed Elsevier journal articles*.

Research Projects and Grants

1. KEY PERSONNEL, “[Collaborative Research: VINES Track 2: AgSlicing: Predictable RAN and Spectrum Slicing for Precision Agriculture](#),” **NSF VINES**, (*Total Funding: \$4,651,625*), Iowa State University (lead), University of Virginia, Skylark Wireless, Salin 247, Federated Wireless, Sep 2026 – Aug 2029
2. RESEARCHER, “5G Sidelink Evaluation,” **John Deere**, (*Total Funding: \$112,000*), Nov 2025 – Oct 2026
3. RESEARCHER, “Ara4Rural: ARA Infrastructure and Community Building for Rural Wireless Applications,” **NSF NeTS**, (*Total Funding: \$1,000,000*), Iowa State University, Dec 2024 – Nov 2026
4. GRADUATE RESEARCHER, “[Acceleration of Compatibility and Commercialization for Open RAN Deployments \(ACCoRD\)](#),” **NTIA Public Wireless Supply Chain Innovation Fund**, (*Total Funding: \$42,299,693; ISU Share: \$1,000,000*), AT&T, Verizon, University of Texas at Dallas, Northeastern University, Iowa State University, Rutgers University, Virginia Tech, NTT DOCOMO, Reliance Jio, Jan 2024 – Dec 2027
5. GRADUATE RESEARCHER, “[Open RAN Research, Testing, and Integration for Rural Industries and Communities](#),” **NSF NeTS and TIP**, (*Total Funding: \$1,400,000*), Iowa State University, May 2023 – Apr 2025

6. GRADUATE RESEARCHER, “[Collaborative Research: SII-NRDZ: ARA-NRDZ: From Site and Application Investigation to Prototyping and Field Testing](#),” **NSF NRDZ**, (*Total Funding: \$500,000; ISU Share: \$485,000*), Iowa State University, Stony Brook University, Oct 2022 – Sep 2026
7. LEAD 5G AND OPEN RAN DEVELOPER, “[ARA Wireless Living Lab for Smart and Connected Rural Communities](#),” **NSF PAWR**, (*Total Funding: \$16,000,000*), Iowa State University, University of California Irvine, Ohio State University, International Computer Science Institute, Jun 2021 – May 2026

Awards and Honors

1. **Best Demo Award**, [Midscale Experimental Research Infrastructure Forum \(MERIF 2026\)](#) **Sep 2026**
At-Scale, Real-World Physical AI for Precision Agriculture on the ARA Wireless Living Lab Washington, DC, USA
2. **Travel Grant**, [Midscale Experimental Research Infrastructure Forum \(MERIF 2026\)](#) **Aug 2026**
The George Washington University Washington, DC, USA
3. **Best Paper Award**, [11th IEEE International Conference on Network Softwarization \(IEEE NetSoft 2025\)](#) **Jun 2025**
AraRACH: Enhancing NextG Random Access Reliability in Programmable Wireless Living Labs Budapest, Hungary
4. **Best Demo Award**, [Midscale Experimental Research Infrastructure Forum \(MERIF 2024\)](#) **Sep 2024**
ARA PAWR: Enabling Wireless Experiments with Programmable COTS RAN and x-Haul Platforms Kansas City, MO, USA
5. **Best Paper Award**, [17th ACM Workshop on Wireless Network Testbeds, Experimental Evaluation & Characterization \(ACM WINTech 2023\)](#) **Oct 2023**
AraMIMO: Programmable TVWS mMIMO Living Lab for Rural Wireless Madrid, Spain
6. **Takano Electrical Engineering Scholarship**, Iowa State University **May 2021**
7. **Valedictorian (Overall Best Graduating Student)**, KNUST **Jun 2019**
8. **Best Graduating Student in Telecommunication Engineering**, Vodafone Ghana **Mar 2019**
9. **First Runner-Up**, Huawei ICT Competition (Routing, Switching, and Wireless LANs), Sandton, South Africa **Feb 2019**
10. **Huawei Certified Networking Associate (HCNA)**, Huawei **Dec 2018**
11. **National Union of Ghana Students Award for Overall Best Engineering Student**, KNUST **May 2018**
12. **Provost Award for Overall Best Student in Telecommunication Engineering**, KNUST **Oct 2016 – Oct 2019**
13. **Provost Award for Excellent Student in Telecommunication Engineering**, KNUST **Oct 2016 – Oct 2019**

Publications

Journal Articles

1. T. U. Islam, **J. O. Boateng**, M. Nadim, G. Zu, M. Shahid, X. Li, T. Zhang, S. Reddy, W. Xu, A. Atalar, V. Lee, Y. Chen, E. Gossling, E. Permatasari, C. Somiah, O. Perrin, Z. Meng, R. Afzal, S. Babu, M. Soliman, A. Hussain, D. Qiao, M. Zheng, O. Boyraz, Y. Guan, A. Arora, M. Y. Selim, A. Ahmad, M. B. Cohen, M. Luby, R. Chandra, J. Gross, K. Keahey, and H. Zhang, “[Design and Implementation of ARA Wireless Living Lab for Rural Broadband and Applications](#),” *Elsevier Computer Networks*, vol. 263, p. 111188, 2025.
2. J. J. Kponyo, J. O. Agyemang, G. S. Klogo, and **J. O. Boateng**, “[Lightweight and Host-Based Denial of Service \(DoS\) Detection and Defense Mechanism for Resource-Constrained IoT Devices](#),” *Elsevier Internet of Things*, vol. 12, p. 100319, 2020.
3. J. O. Agyemang, J. J. Kponyo, G. S. Klogo, and **J. O. Boateng**, “[Lightweight Rogue Access Point Detection Algorithm for WiFi-Enabled Internet of Things \(IoT\) Devices](#),” *Elsevier Internet of Things*, vol. 11, p. 100200, 2020.

Conference Papers

1. G. Zu, **J. O. Boateng**, V. S. Advani, T. U. Islam, V. Lee, S. Babu, M. Nadim, D. Qiao, M. Y. Selim, and H. Zhang, “Real-World Integration and Evaluation of Open-Source 5G Core with Commercial RAN,” in *Proc. IEEE Military Communications Conference (IEEE MILCOM)*, Los Angeles, CA, USA, Oct. 2025, pp. 1296–1301.
2. **J. O. Boateng**, T. Zhang, G. Zu, T. U. Islam, S. Babu, F. Kaltenberger, R. Schmidt, H. Zhang, and D. Qiao, “AraRACH: Enhancing NextG Random Access Reliability in Programmable Wireless Living Labs,” in *Proc. 11th IEEE International Conference on Network Softwarization (IEEE NetSoft)*, Budapest, Hungary, Jun. 2025, pp. 362–370 [**Best Paper Award**].
3. **J. O. Boateng**, T. Zhang, G. Zu, T. U. Islam, S. Babu, H. Zhang, and D. Qiao, “AraSDR: End-to-End, Fully-Programmable Living Lab for 5G and Beyond,” in *Proc. IEEE International Conference on Communications (IEEE ICC)*, Denver, CO, USA, Jun. 2024, pp. 1758–1763.
4. T. Zhang, **J. O. Boateng**, T. U. Islam, A. Ahmad, H. Zhang, and D. Qiao, “Ara-O-RAN: End-to-End Programmable O-RAN Living Lab for Agriculture and Rural Communities,” in *Proc. IEEE INFOCOM Workshops (INFOCOM WKSHPs)*, May 2024, pp. 1–6.
5. T. U. Islam, T. Zhang, **J. O. Boateng**, E. Gossling, G. Zu, S. Babu, H. Zhang, and D. Qiao, “AraMIMO: Programmable TVWS mMIMO Living Lab for Rural Wireless,” in *Proc. 17th ACM Workshop on Wireless Network Testbeds, Experimental Evaluation & Characterization (ACM WiNTECH)*, Madrid, Spain, Oct. 2023, pp. 9–16 [**Best Paper Award**].
6. M. Shahid, S. Babu, H. Zhang, D. Qiao, Y. Guan, **J. O. Boateng**, T. U. Islam, G. Zu, A. Kamal, and M. Zheng, “Wireless Guard for Trustworthy Spectrum Management,” in *Proc. 16th ACM Workshop on Wireless Network Testbeds, Experimental Evaluation & Characterization (ACM WiNTECH)*, Sydney, Australia, Oct. 2022, pp. 32–39.

Demos and Posters

1. T. U. Islam and **J. O. Boateng**, “At-Scale, Real-World Physical AI for Precision Agriculture on the ARA Wireless Living Lab,” in *Midscale Experimental Research Infrastructure Forum (MERIF ’26)*, Washington, DC, USA, Sep. 2026 [**Best Demo Award**].
2. **J. O. Boateng** and H. Zhang, “Multi-Cell srsRAN in ARA Wireless Living Lab: Exploring Distance Effects, Interference Coordination and System-Level Tuning,” in *srsRAN Project Fall Workshop*, Arlington, VA, USA, Nov. 2025.
3. T. Zhang, **J. O. Boateng**, and H. Zhang, “Real-World Experiments with Predictable URLLC Using OAI and ARA Wireless Living Lab,” in *First OpenAirInterface Foundation U.S. Hands-on Workshop*, Austin, TX, USA, Nov. 2025.
4. G. Zu, **J. O. Boateng**, V. S. Advani, T. U. Islam, V. Lee, S. Babu, M. Nadim, D. Qiao, M. Y. Selim, and H. Zhang, “Demo: Real-World Integration and Evaluation of Open-Source 5G Core with Commercial RAN,” in *Proc. IEEE MILCOM*, Los Angeles, CA, USA, Oct. 2025, pp. 881–882.
5. T. U. Islam, **J. O. Boateng**, M. Nadim, G. Zu, M. Shahid, X. Li, T. Zhang, S. Reddy, W. Xu, A. Atalar, V. Lee, Y. F. Chen, E. Gossling, E. Permatasari, Z. Meng, S. Babu, M. Soliman, A. Hussain, D. Qiao, M. Zheng, O. Boyraz, Y. Guan, A. Arora, M. Selim, A. Ahmad, M. B. Cohen, and H. Zhang, “Demo: ARA PAWR Wireless Living Lab for Smart and Connected Rural Communities,” in *Proc. 32nd IEEE International Conference on Network Protocols (IEEE ICNP)*, Charleroi, Belgium, Oct. 2024, pp. 1–3.
6. **J. O. Boateng**, T. Zhang, S. Babu, and H. Zhang, “ARA-Enabled End-to-End 5G and Open RAN Prototyping Using srsRAN and FlexRIC,” in *srsRAN Project Fall Workshop*, Arlington, VA, USA, Oct. 2024.
7. T. U. Islam, M. Nadim, G. Zu, V. Lee, O. J. Perrin, **J. O. Boateng**, M. Shahid, T. Zhang, E. K. A. Permatasari, Z. Meng, S. Babu, D. Qiao, and H. Zhang, “ARA PAWR: Enabling Wireless Experiments with Programmable COTS RAN and x-Haul Platforms,” in *Midscale Experimental Research Infrastructure Forum (MERIF ’24)*, Kansas City, MO, USA, Sep. 2024 [**Best Demo Award**].
8. T. Zhang, Z. Meng, E. Permatasari, **J. O. Boateng**, T. U. Islam, M. Nadim, V. Lee, S. Babu, H. Zhang, and D. Qiao, “ARA-Enabled End-to-End Open-Source Prototyping of 5G/O-RAN Solutions for Predictable Low-Latency Communications,” in *Midscale Experimental Research Infrastructure Forum (MERIF ’24)*, Kansas City, MO, USA, Sep. 2024.

9. **J. O. Boateng**, T. Zhang, Z. Meng, H. Zhang, and D. Qiao, “End-to-End 5G and O-RAN Prototyping in ARA,” in *OpenAirInterface 10th Anniversary Workshop*, Biot, France, Sep. 2024.
10. T. U. Islam, **J. O. Boateng**, G. Zu, M. Shahid, M. Nadim, W. Xu, T. Zhang, S. Reddy, X. Li, A. Atalar, Y. Chen, S. Babu, H. Zhang, D. Qiao, M. Zheng, Y. Guan, O. Boyraz, A. Arora, M. Selim, and M. B. Cohen, “[ARA PAWR: Wireless Living Lab for Smart and Connected Rural Communities](#),” in *Proc. 29th Annual International Conference on Mobile Computing and Networking (ACM MobiCom)*, Madrid, Spain, Oct. 2023, Article 98, pp. 1–3.
11. **J. O. Boateng**, T. Zhang, H. Zhang, and D. Qiao, “ARA Wireless Living Lab and End-to-End Open-Source 5G Experiment with srsRAN,” in *srsRAN Project Fall Workshop*, Arlington, VA, USA, Oct. 2023.
12. **J. O. Boateng**, T. Zhang, S. Babu, T. U. Islam, M. Shahid, and H. Zhang, “ARA Rural Wireless Living Lab and End-to-End OTA Containerized OAI 5G-NR Support,” in *Fall 2022 OpenAirInterface North American Workshop*, San Diego, CA, USA, Nov. 2022.

Professional Service

Technical Program Committee (TPC) Member

- Workshop on Open Research Infrastructures and Toolkits for 6G (OpenRIT6G) @ IEEE WCNC 2027
- ACM Workshop on Wireless Network Testbeds, Experimental Evaluation & Characterization (ACM WiNTECH) 2026
- IEEE Military Communications Conference – Demos Track (IEEE MILCOM) 2026

Organizing Committees

- Demo and Poster Chair, [AgWireless’26: Accelerating Concerted 6G, AgTech & Multi-Use Innovation](#), Ames, IA 2026
- Organizer, NSF AgRuralG Workshop and ARA User Program & Local Organizing Committee, Ames, IA 2026

Journal Reviewer

- IEEE Transactions on Wireless Communications (IEEE TWC) 2023 – 2026
- IEEE Access 2026
- Journal of Communications and Networks (JCN) 2026

Conference Reviewer

- IEEE Conference on Computer Communications (IEEE INFOCOM) 2022 – 2025
- IEEE Military Communications Conference (IEEE MILCOM) 2025 – 2026
- ACM Workshop on Wireless Network Testbeds, Experimental Evaluation & Characterization (ACM WiNTECH) 2026

Industry Alliances and Professional Memberships

- [AI-RAN Alliance](#) – Member, Working Group 2 (AI-and-RAN) 2026 – Present
- [OCUDU Project \(Linux Foundation\)](#) – Technical Steering Committee (TSC) Working Groups
 - Member, AI-RAN Working Group 2026 – Present
 - Member, Hardware-Acceleration Working Group 2026 – Present
- IEEE Communications Society (ComSoc) 2024 – Present

Teaching and Mentoring

Iowa State University

- Graduate Student Mentor, NSF Research Experiences for Undergraduates (REU) 2024, 2025

Kwame Nkrumah University of Science and Technology

- EE 472 – Digital Signal Processing Aug 2019 – Sep 2020
- TE 461 – Computer Application and Project Design Aug 2019 – Sep 2020
- COE 475 – Computer Networking Aug 2019 – Sep 2020

Invited Talks and Workshop Presentations

- MERIF 2026 Demo and Poster Night, *Washington, DC, USA* **Sep 2026**
- srsRAN Project Fall Workshop, *Arlington, VA, USA* **Nov 2025**
- First OAI Foundation U.S. Hands-on Workshop, *Austin, TX, USA* **Nov 2025**
- srsRAN Project Fall Workshop, *Arlington, VA, USA* **Oct 2024**
- OpenAirInterface 10th Anniversary Workshop, *Biot, France* **Sep 2024**
- Joint OSC/OSFG-OAI Workshop: End-to-End Reference Designs for O-RAN, *Burlington, MA, USA* **Nov 2023**
- srsRAN Project Fall Workshop, *Arlington, VA, USA* **Oct 2023**
- Fall 2022 OpenAirInterface North American Workshop, *San Diego, CA, USA* [\[Video\]](#) **Nov 2022**
- CHI@Edge Workshop: Applications and Network Measurements for SDR on CHI@Edge, *Online* [\[Video\]](#) **Oct 2021**

Technical Skills

GPU-Accelerated and AI-RAN: [NVIDIA Aerial CUDA-Accelerated RAN](#) (cuPHY L1), CUDA-accelerated OCUDU PHY, SCF FAPI (L1/L2 interface), NVIDIA GPU platforms, AI-RAN (AI-for-RAN, AI-on-RAN)

5G/6G RAN Stacks and Core: [OCUDU](#) (formerly srsRAN), [Duranta](#) (formerly OpenAirInterface (OAI)), Open5GS, SD-Core, SD-RAN.

O-RAN and RAN Control: FlexRIC near-RT RIC, E2 interface, xApps, RAN slicing, 5G NR SA (TDD), random access (RACH), NTN, O-RAN fronthaul splits 7.2x and 8

Software-Defined Radio and RF: USRP X410, X310, N320, B210; custom PA/LNA RF front-ends; GNU Radio

Programming: Python, C, C++, MATLAB, Bash

Systems, Virtualization, and Cloud: Linux, Docker, Kubernetes (K8s), KVM, Git, GitLab

Monitoring and Measurement: Grafana, InfluxDB, Telegraf, GNUPlot

Documentation and Prototyping: $\text{\LaTeX}$, Sphinx, Autodesk Fusion 360, Ultimaker Cura (3D printing)